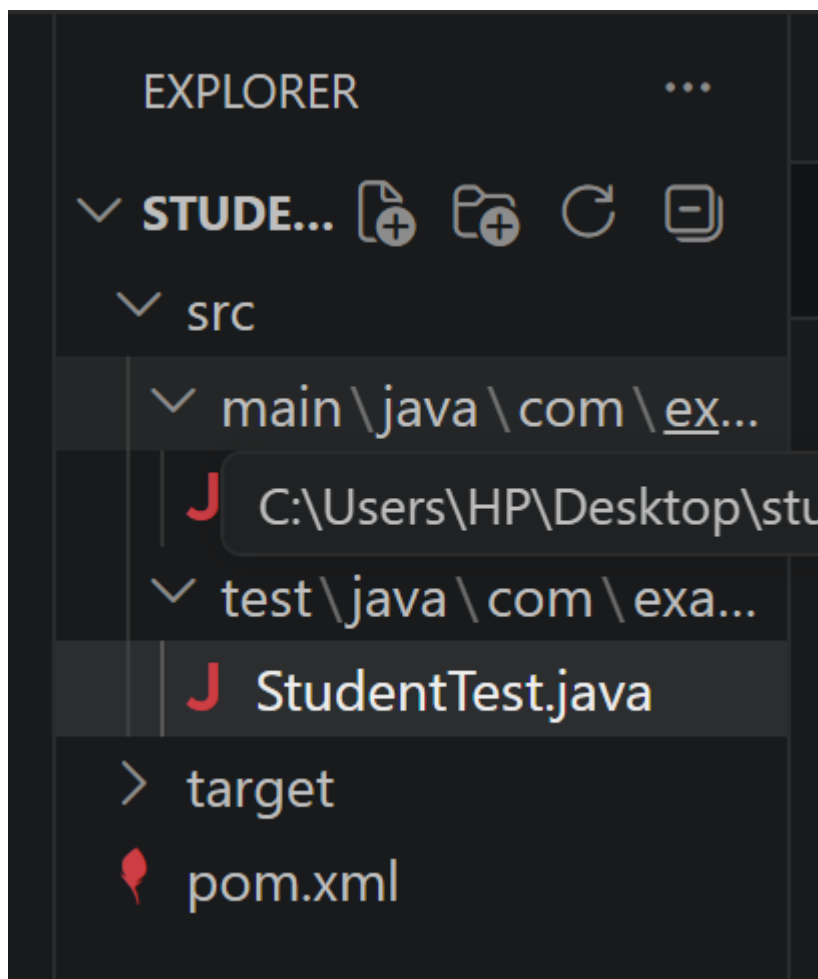


Agile lab fat

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REG NO:23MIS0585

FILE STRUCTURE:



STUDENTAPP.JAVA

```
package com.example;
```

```
import java.util.*;
```

```

class Student {
    String name;
    int cat, assignment, fat;
    int total;
    double gpa;
    String grade;

    Student(String name, int cat, int assignment, int fat) {
        this.name = name;
        this.cat = cat;
        this.assignment = assignment;
        this.fat = fat;
        calculate();
    }

    void calculate() {
        total = cat + assignment + fat;
        gpa = total / 10.0;

        if (total >= 90) grade = "A";
        else if (total >= 75) grade = "B";
        else if (total >= 60) grade = "C";
        else grade = "D";
    }
}

public class StudentApp {

    public static Student findTopper(List<Student> students) {
        return Collections.max(students, Comparator.comparingInt(s -> s.total));
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        List<Student> students = new ArrayList<>();

        for (int i = 0; i < 5; i++) {
            System.out.println("\nEnter details for Student " + (i + 1));

            System.out.print("Name: ");
            String name = sc.next();

            System.out.print("CAT (30): ");
            int cat = sc.nextInt();

            System.out.print("Assignment (20): ");

```

```

        int assignment = sc.nextInt();

        System.out.print("FAT (50): ");
        int fat = sc.nextInt();

        students.add(new Student(name, cat, assignment, fat));
    }

    Student topper = findTopper(students);

    System.out.println("\n===== STUDENT RESULTS =====");
    for (Student s : students) {
        System.out.println(
            s.name + " | Total: " + s.total +
            " | GPA: " + s.gpa +
            " | Grade: " + s.grade
        );
    }

    System.out.println("\n Topper: " + topper.name + " (" + topper.total + ")");
}
}

```

```

> main > java > com > example > J StudentApp.java
1  package com.example;
2
3  import java.util.*;
4
5  class Student {
6      String name;
7      int cat, assignment, fat;
8      int total;
9      double gpa;
10     String grade;
11
12     Student(String name, int cat, int assignment, int fat) {
13         this.name = name;
14         this.cat = cat;
15         this.assignment = assignment;
16         this.fat = fat;
17         calculate();
18     }
19
20     void calculate() {
21         total = cat + assignment + fat;
22         gpa = total / 10.0;
23
24         if (total >= 90) grade = "A";
25         else if (total >= 75) grade = "B";
26         else if (total >= 60) grade = "C";
27         else grade = "D";
28     }
29 }
30
31 public class StudentApp {
32
33     public static Student findTopper(List<Student> students) {

```

STUDENTTEST.JAVA

```
package com.example;
```

```
import org.junit.Test;
import static org.junit.Assert.*;
import java.util.*;
```

```
public class StudentTest {
```

```
    @Test
```

```

public void testTopperCalculation() {
    List<Student> students = new ArrayList<>();

    students.add(new Student("A", 25, 15, 40)); // 80
    students.add(new Student("B", 30, 20, 50)); // 100
    students.add(new Student("C", 20, 10, 30)); // 60

    Student topper = StudentApp.findTopper(students);

    assertEquals("B", topper.name);
}
}

```

```

src > test > java > com > example > J StudentTest.java
1  package com.example;
2
3  import org.junit.Test;
4  import static org.junit.Assert.*;
5  import java.util.*;
6
7  public class StudentTest {
8
9      @Test
10     public void testTopperCalculation() {
11         List<Student> students = new ArrayList<>();
12
13         students.add(new Student("A", 25, 15, 40)); // 80
14         students.add(new Student("B", 30, 20, 50)); // 100
15         students.add(new Student("C", 20, 10, 30)); // 60
16
17         Student topper = StudentApp.findTopper(students);
18
19         assertEquals("B", topper.name);
20     }
21 }

```

POM.XML

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.example</groupId>
  <artifactId>student-performance-system</artifactId>
  <version>1.0-SNAPSHOT</version>

  <dependencies>
    <!-- JUnit for testing -->
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>4.13.2</version>
      <scope>test</scope>
    </dependency>
  </dependencies>

  <build>
    <plugins>
      <!-- Compiler Plugin -->
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-compiler-plugin</artifactId>
        <version>3.8.1</version>
        <configuration>
          <source>1.8</source>
          <target>1.8</target>
        </configuration>
      </plugin>
    </plugins>
  </build>

</project>
```

```

pom.xml
1  <project xmlns="http://maven.apache.org/POM/4.0.0"
2      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3      xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
4      http://maven.apache.org/xsd/maven-4.0.0.xsd">
5
6      <modelVersion>4.0.0</modelVersion>
7
8      <groupId>com.example</groupId>
9      <artifactId>student-performance-system</artifactId>
10     <version>1.0-SNAPSHOT</version>
11
12     <dependencies>
13         <!-- JUnit for testing -->
14         <dependency>
15             <groupId>junit</groupId>
16             <artifactId>junit</artifactId>
17             <version>4.13.2</version>
18             <scope>test</scope>
19         </dependency>
20     </dependencies>
21
22     <build>
23         <plugins>
24             <!-- Compiler Plugin -->
25             <plugin>
26                 <groupId>org.apache.maven.plugins</groupId>
27                 <artifactId>maven-compiler-plugin</artifactId>
28                 <version>3.8.1</version>
29                 <configuration>
30                     <source>1.8</source>
31                     <target>1.8</target>
32                 </configuration>
33             </plugin>

```

MVN CLEAN PACKAGE

```

student-performance-system> mvn clean package

```

```

[INFO] --- compiler:3.8.1:testCompile (default-testCompile) @ student-performance-system ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to C:\Users\HP\Desktop\student-performance-system\target\test-classes
[INFO] --- surefire:3.5.4:test (default-test) @ student-performance-system ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.StudentTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.164 s -- in com.example.StudentTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO] --- jar:3.5.0:jar (default-jar) @ student-performance-system ---
[INFO] Building jar: C:\Users\HP\Desktop\student-performance-system\target\student-performance-system-1.0-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.101 s

```

Git comments:

```

\student-performance-system> git init
\student-performance-system> git add .

```

```

student-performance-system> git commit -m "first commieesset"

mmieesset
ertions(+), 5 deletions(-)
student-performance-system> git remote add origin https://github.com/vishnuu45/student-performace-system.git
ready exists.
student-performance-system> git push -u origin main

```

```

Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 853 bytes | 170.00 KiB/s, done.
Total 7 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 5 local objects.
To https://github.com/vishnuu45/student-performace-system.git
 7414198..f2ef191 main -> main
branch 'main' set up to track 'origin/main'.

```


vishnuu45 / student-performace-system

Q

Type [7] to

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security and quality

Insights

Settings

student-performace-system

Public

Pin

Watch 0

main

1 Branch

0 Tags

Go to file

t

Add file

<> Code

src	first commieeet	13 minutes ago
target	first commieeset	2 minutes ago
pom.xml	first commieeet	13 minutes ago

Piple script:

```
pipeline {
  agent any

  tools {
    maven 'Maven3'
  }

  stages {

    stage('Clean Workspace') {
      steps {
        deleteDir()
      }
    }

    stage('Clone Repository') {
      steps {
        git branch: 'main',
          url: 'https://github.com/vishnuu45/student-performace-system.git',
          credentialsId: 'github-cred'
      }
    }

    stage('Build Project') {
      steps {
        bat 'mvn clean package'
```

```

    }
}

stage('Run Tests') {
    steps {
        bat 'mvn test'
    }
}

stage('Archive JAR') {
    steps {
        archiveArtifacts artifacts: 'target/*.jar', fingerprint: true
    }
}

stage('Success Message') {
    steps {
        echo 'CI/CD Pipeline Executed Successfully!'
    }
}

}

post {
    success {
        echo 'Build SUCCESS'
    }
    failure {
        echo 'Build FAILED'
    }
}
}

```

Pipeline script

Script ?

```

1 pipeline {
2     agent any
3
4     tools {
5         maven 'Maven3'
6     }
7
8     stages {
9
10        stage('Clean Workspace') {
11            steps {
12                deleteDir()
13            }
14        }
15    }

```

try sample Pipeline...

☒ Use Groovy Sandbox ?

Script ?

```
15
16 stage('Clone Repository') {
17     steps {
18         git branch: 'main',
19             url: 'https://github.com/vishnuu45/student-performace-system.git',
20             credentialsId: 'github-cred'
21     }
22 }
23
24 stage('Build Project') {
25     steps {
26         bat 'mvn clean package'
27     }
28 }
29
```

try sample Pipeline... ▼

☒ Use Groovy Sandbox ?

Script ?

```
29
30 stage('Run Tests') {
31     steps {
32         bat 'mvn test'
33     }
34 }
35
36 stage('Archive JAR') {
37     steps {
38         archiveArtifacts artifacts: 'target/*.jar', fingerprint: true
39     }
40 }
41
42 stage('Success Message') {
43     steps {
44         echo 'CI/CD Pipeline Executed Successfully!'
```

try sample Pipeline... ▼

☒ Use Groovy Sandbox ?

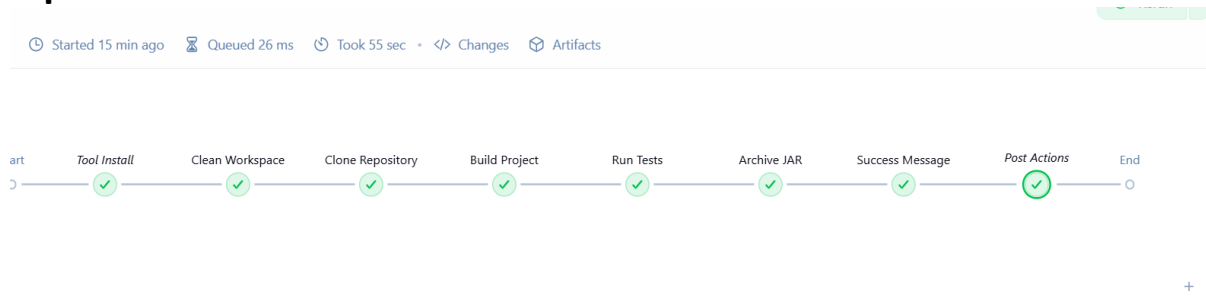
```
43     steps {
44         echo 'CI/CD Pipeline Executed Successfully!'
45     }
46 }
47 }
48
49 post {
50     success {
51         echo 'Build SUCCESS'
52     }
53     failure {
54         echo 'Build FAILED'
55     }
56 }
57 }
```

try sample Pipeline... ▼

```
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ student-performance-system ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -----
[INFO]  T E S T S
[INFO] -----
[INFO] Running com.example.StudentTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.211 s -- in com.example.StudentTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time:  6.786 s
[INFO] Finished at: 2026-04-13T10:27:18+05:30
[INFO] -----
```

```
[Pipeline] { (Success Message)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] echo
✅ CI/CD Pipeline Executed Successfully!
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
🎉 Build SUCCESS
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Pipeline overview:



Q Search



- ✓ Tool Install 0.31s
- ✓ Clean Workspace 0.47s
- ✓ Clone Repository 3s
- ✓ Build Project 37s
- ✓ Run Tests 10s
- ✓ Archive JAR 0.36s
- ✓ Success Message 0.25s
- ✓ Post Actions 75ms

✓ Build Project

🕒 37s

🕒 Started 16m ago

💻 Jenkins



✓ Use a tool from a predefined Tool Installation [Maven3](#) >

39ms



✓ Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step. >

66ms



✓ Windows Batch Script [mvn clean package](#) >

37s



